# T1 — Atomic Structure, Bonding and Energetics

## 1.1 Atomic Models and Fundamental Particles

The atom consists of a dense, positively charged nucleus containing protons and neutrons, surrounded by electrons in quantised energy levels (shells and sub-shells). Understanding of atomic structure evolved through several models:

- **Dalton (1803):** Atoms are indivisible solid spheres. Different elements have different atoms.
- **Thomson (1897):** Discovered the electron. Proposed the "plum pudding" model — electrons embedded in a sphere of positive charge.
- **Rutherford (1911):** Alpha particle scattering experiment showed most of the atom is empty space, with a tiny dense positive nucleus. Most alpha particles passed straight through gold foil; a few were deflected at large angles; very few bounced back.
- **Bohr (1913):** Electrons orbit the nucleus in fixed energy levels (shells). Electrons can jump between levels by absorbing or emitting photons of specific energy. This explained atomic emission spectra.
- **Modern quantum model:** Electrons occupy orbitals (regions of space where there is a high probability of finding an electron), not fixed orbits. Orbitals have characteristic shapes and energies.

**Mass number (A):** The total number of protons and neutrons in the nucleus.
**Atomic number (Z):** The number of protons in the nucleus (defines the element).
**Isotopes:** Atoms of the same element with the same number of protons but different numbers of neutrons. They have identical chemical properties (same electron configuration) but different physical properties (different mass).

| Particle | Relative Mass | Relative Charge | Location |
|---|---|---|---|
| Proton | 1 | +1 | Nucleus |
| Neutron | 1 | 0 | Nucleus |

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| | | | |
|---|---|---|---|
| Electron | 1/1836 (≈ 0.0005) | -1 | Orbitals around the nucleus |

## Relative Atomic Mass and Mass Spectrometry

**Relative atomic mass ($A_r$):** The weighted mean mass of an atom of an element, relative to 1/12th the mass of a carbon-12 atom. It is dimensionless (no units).

**Relative isotopic mass:** The mass of one atom of an isotope relative to 1/12th the mass of a carbon-12 atom.

**Relative molecular mass ($M_r$):** The sum of the relative atomic masses of all atoms in a molecular formula.

$$A_r = \Sigma(\text{isotopic mass} \times \text{percentage abundance}) / 100$$

**Mass spectrometry** is used to determine isotopic masses and their relative abundances. The key stages of a time-of-flight (TOF) mass spectrometer are:

1. **Ionisation:** The sample is vaporised and ionised by electron bombardment (electron impact) or by electrospray ionisation (ESI). Electron impact: $M(g) \rightarrow M^+(g) + e^-$. Electrospray: the sample is dissolved in a volatile solvent and sprayed through a charged needle, gaining a proton: $M + H^+ \rightarrow MH^+$.
2. **Acceleration:** Ions are accelerated by an electric field. All ions gain the same kinetic energy ($KE = ½mv^2$), so lighter ions travel faster.
3. **Drift (flight tube):** Ions separate in a field-free region based on their mass-to-charge ratio (m/z). Lighter ions reach the detector first.
4. **Detection:** Ions hit the detector, generating a current proportional to their abundance.

The mass spectrum shows peaks at each m/z value, with peak height proportional to abundance. For elements, this allows calculation of $A_r$. For molecules, the molecular ion peak ($M^+$) gives the relative molecular mass.

**Exam Tip:** When calculating $A_r$ from a mass spectrum, check whether abundances are given as percentages or as raw peak heights. If raw heights, convert to percentages first by dividing each by

the total sum. Example: Boron has two isotopes: $^{10}B$ (20%) and $^{11}B$ (80%). $A_r$ = (10 × 20 + 11 × 80) / 100 = 10.8. Always give your answer to 1 decimal place unless told otherwise.

## 1.2 Electron Configuration

Electrons occupy atomic orbitals — regions of space where there is a high probability of finding an electron. Each orbital can hold a maximum of two electrons with opposite spins (the **Pauli exclusion principle**).

### Sub-shells and Orbitals

| Sub-shell | Number of Orbitals | Maximum Electrons | Shape |
|---|---|---|---|
| s | 1 | 2 | Spherical |
| p | 3 | 6 | Dumbbell (three orientations: $p_x$, $p_y$, $p_z$) |
| d | 5 | 10 | Cloverleaf (various orientations) |
| f | 7 | 14 | Complex shapes |

**Three rules govern how electrons fill orbitals:**

1. **Aufbau principle:** Electrons fill the lowest energy sub-shell available first. The filling order is: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p...
2. **Pauli exclusion principle:** Each orbital holds a maximum of two electrons, which must have opposite spins (represented by ↑↓).
3. **Hund's rule:** Electrons occupy orbitals of the same sub-shell singly (with parallel spins) before pairing up. This minimises electron-electron repulsion.

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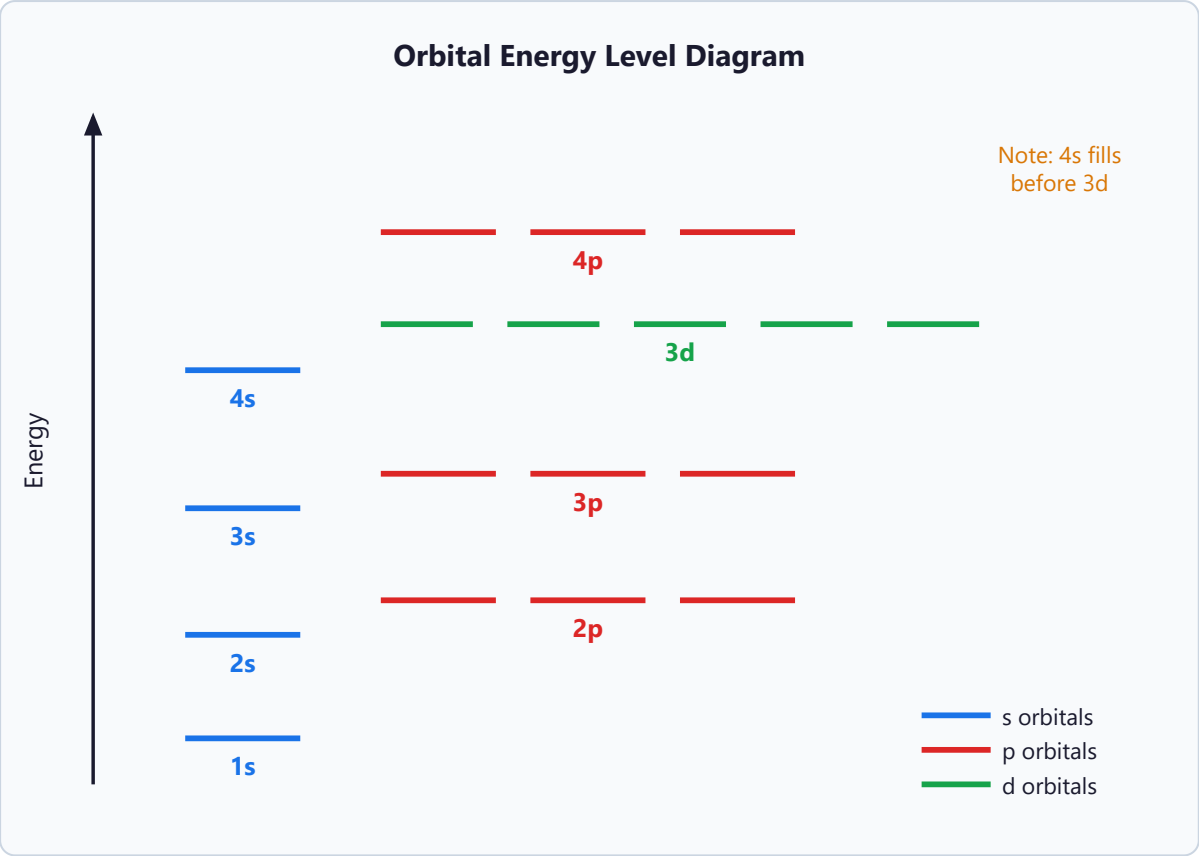



## Electron Configuration Notation

**Full notation** writes out every sub-shell, e.g. iron (Z=26): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$.

**Shorthand notation** uses the previous noble gas core: [Ar] $3d^6\ 4s^2$.

**Key anomalous configurations:**

- **Chromium:** [Ar] $3d^5\ 4s^1$ (NOT $3d^4\ 4s^2$) — a half-filled 3d sub-shell provides extra stability due to exchange energy (electrons with parallel spins in different orbitals have lower repulsion).
- **Copper:** [Ar] $3d^{10}\ 4s^1$ (NOT $3d^9\ 4s^2$) — a fully filled 3d sub-shell is more stable.

**Ions of transition metals:** When forming ions, electrons are removed from the 4s sub-shell first (not 3d), because in the presence of d electrons, 4s is actually at a higher energy than 3d. Examples:

- $Fe^{2+}$: [Ar] $3d^6$ (both 4s electrons removed)
- $Fe^{3+}$: [Ar] $3d^5$ (both 4s electrons and one 3d electron removed)
- $Cu^{2+}$: [Ar] $3d^9$

**Exam Tip:** The anomalous configurations of Cr and Cu are very commonly examined. Always remove 4s electrons first when writing ion configurations. A common mistake is writing $Fe^{2+}$ as [Ar] $3d^4$ $4s^2$ — this is wrong. The correct answer is [Ar] $3d^6$.

## 1.3 Ionisation Energies

**First ionisation energy ($IE_1$):** The energy required to remove one mole of the most loosely held electrons from one mole of gaseous atoms to form one mole of gaseous 1+ ions.

$M(g) \rightarrow M^+(g) + e^-$ [always endothermic]

**Second ionisation energy ($IE_2$):** The energy to remove a second electron from the gaseous 1+ ion:

$M^+(g) \rightarrow M^{2+}(g) + e^-$

### Factors Affecting Ionisation Energy

- **Nuclear charge (Z):** More protons = stronger attraction on outer electrons = higher IE
- **Atomic radius/distance:** Greater distance from nucleus = weaker electrostatic attraction = lower IE
- **Electron shielding:** More complete inner shells between the nucleus and the outer electron = reduced effective nuclear charge = lower IE
- **Sub-shell structure:** Electrons in p orbitals are slightly higher in energy (easier to remove) than electrons in s orbitals of the same shell. Paired electrons experience extra repulsion, making them easier to remove than unpaired electrons.

### Trends in First Ionisation Energy

**Across a period (e.g. Na to Ar):** General increase due to increasing nuclear charge with the same number of inner shielding electrons. However, two notable decreases occur:

1. **Group 2 → Group 3 (Mg → Al):** Al ($1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^1$) loses a 3p electron, which is higher in energy and further from the nucleus than the 3s electron lost by Mg. The 3p electron is also shielded by the $3s^2$ pair.
2. **Group 5 → Group 6 (N → O / P → S):** In Group 6, the fourth electron in the 3p sub-shell must pair up with an existing electron. The electron-electron repulsion in this pair makes the paired electron easier

to remove.

**Down a group:** Ionisation energy decreases because: (1) atomic radius increases (more shells), (2) shielding by inner electrons increases. Both effects outweigh the increase in nuclear charge.

### Successive Ionisation Energies

Each successive ionisation energy is always higher than the previous one because: (1) the ion becomes more positive, so there is greater electrostatic attraction on the remaining electrons, and (2) there are fewer electrons repelling each other, so each remaining electron is held more tightly.

A **large jump** between consecutive ionisation energies indicates that an electron has been removed from a new, inner shell (which is closer to the nucleus and has less shielding). The position of this jump reveals the group number of the element.

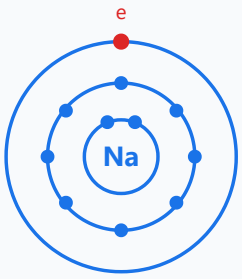
**Exam Tip:** If asked to deduce the group of an unknown element from successive IE data, look for the biggest jump between consecutive IEs. If the big jump is between the 2nd and 3rd IE, the element is in Group 2. If between the 3rd and 4th, it is in Group 3. The number of electrons removed before the big jump equals the number of outer-shell electrons.

## 1.4 Ionic Bonding

**Ionic bonding:** The strong electrostatic attraction between oppositely charged ions. Ionic bonds form when electrons are transferred from a metal atom (forming a cation, $M^{n+}$) to a non-metal atom (forming an anion, $X^{n-}$).

Ionic compounds form **giant ionic lattices** — three-dimensional structures where each ion is surrounded by ions of the opposite charge in a regular repeating pattern. The compound has no overall charge because the total positive charge equals the total negative charge.

**Ionic Bonding: Electron Transfer (Na + Cl)**




$^-$ electron transfer Cl 2, 8, 1 2, 8, 7 $Na^+$ [2,8] $Cl^-$ [2,8,8] Na loses $1e^-$ Cl gains $1e^-$ Strong electrostatic attraction between $Na^+$ and $Cl^-$

## Properties of Ionic Compounds

| Property | Explanation |
|---|---|
| High melting and boiling points | Large amounts of energy needed to overcome the strong electrostatic attractions between many ions throughout the giant lattice |
| Conduct electricity when molten or dissolved in water | Ions are free to move and carry charge |
| Do not conduct electricity when solid | Ions are held in fixed positions in the lattice and cannot move |
| Often soluble in water | Polar water molecules can surround and stabilise individual ions (hydration), releasing enough energy to overcome lattice energy |
| Brittle | A small displacement causes layers of like-charged ions to align, creating strong repulsion that shatters the crystal |

**Factors affecting lattice energy (strength of ionic bonding):**

- **Ionic charge:** Higher charges produce stronger attraction (e.g. MgO has a much higher melting point than NaCl because $Mg^{2+}/O^{2-}$ have double the charges of $Na^+/Cl^-$)

- **Ionic radius:** Smaller ions bring charges closer together, increasing electrostatic attraction (e.g. LiF has a higher lattice energy than NaCl)

## 1.5 Covalent Bonding

**Covalent bond:** A shared pair of electrons between two atoms. The electrostatic attraction between the shared pair of electrons and the nuclei of both atoms holds the atoms together. Covalent bonds form between non-metal atoms.

### Types of Covalent Bond

- **Single bond (σ):** One shared pair of electrons. Formed by head-on overlap of atomic orbitals.
- **Double bond (σ + π):** Two shared pairs. The σ bond is from head-on overlap; the π bond is from sideways overlap of p orbitals.
- **Triple bond (σ + 2π):** Three shared pairs (e.g. N≡N in $N_2$).
- **Dative (coordinate) covalent bond:** Both electrons in the shared pair come from one atom. The donor atom must have a lone pair; the acceptor atom must have an empty orbital. Example: $NH_4^+$ ($NH_3$ donates a lone pair to $H^+$), CO (carbon monoxide has a dative bond from O to C), $[Al(H_2O)_6]^{3+}$.

**Bond strength and length:** Triple bonds are stronger and shorter than double bonds, which are stronger and shorter than single bonds. For example: C-C = 346 kJ/mol (0.154 nm), C=C = 614 kJ/mol (0.134 nm), C≡C = 839 kJ/mol (0.120 nm).

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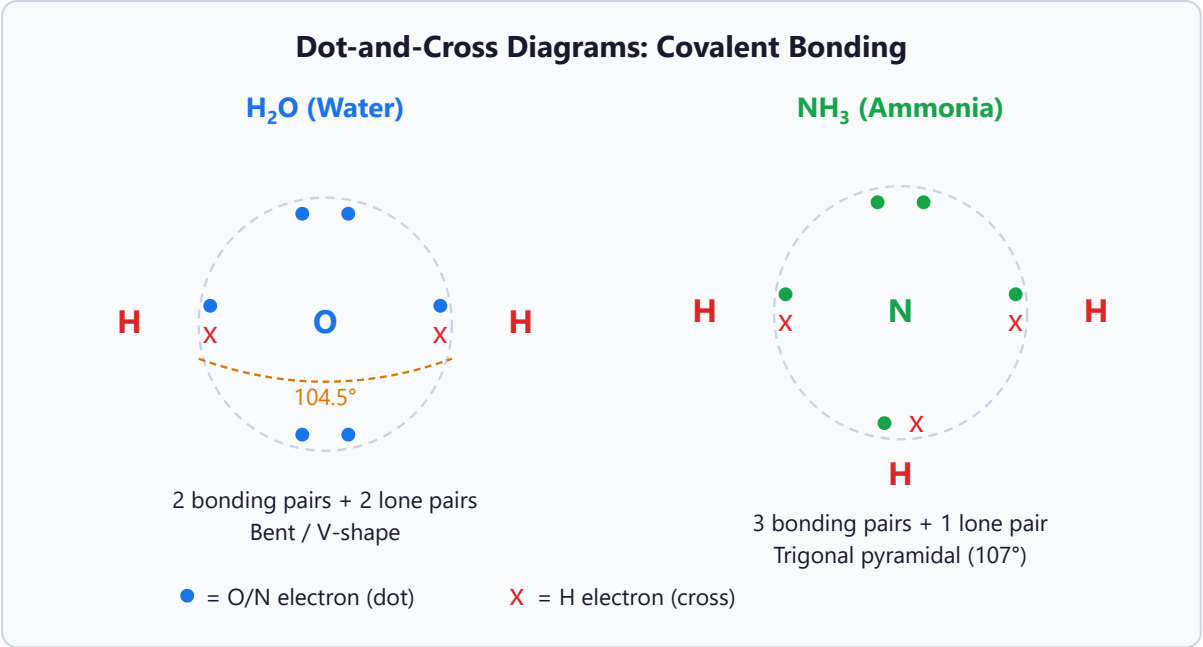


## Shapes of Molecules — VSEPR Theory

**VSEPR (Valence Shell Electron Pair Repulsion) theory:** Electron pairs (both bonding and lone pairs) around a central atom arrange themselves as far apart as possible to minimise electrostatic repulsion between them. The arrangement determines the shape and bond angles of the molecule.

**Repulsion strength order:** lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair. This is because lone pairs are held closer to the central atom (not shared with another nucleus) and occupy more space.

| Total Electron Pairs | Bonding Pairs | Lone Pairs | Shape | Bond Angle | Example |
|---|---|---|---|---|---|
| 2 | 2 | 0 | Linear | 180° | $CO_2$, $BeCl_2$, HCN |
| 3 | 3 | 0 | Trigonal planar | 120° | $BF_3$, $AlCl_3$, $NO_3^-$ |
| 3 | 2 | 1 | Bent / V-shape | ~117-118° | $SO_2$, $NO_2^-$ |
| 4 | 4 | 0 | Tetrahedral | 109.5° | $CH_4$, $NH_4^+$, $SiCl_4$ |
| 4 | 3 | 1 | Trigonal pyramidal | 107° | $NH_3$, $PCl_3$, $H_3O^+$ |

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| | | | | | |
|---|---|---|---|---|---|
| 4 | 2 | 2 | Bent / V-shape | 104.5° | $H_2O$, $H_2S$ |
| 5 | 5 | 0 | Trigonal bipyramidal | 90° and 120° | $PCl_5$, $PF_5$ |
| 6 | 6 | 0 | Octahedral | 90° | $SF_6$, $[Fe(H_2O)_6]^{2+}$ |

**Exam Tip:** When predicting shapes: (1) Draw the Lewis structure. (2) Count all electron pairs around the central atom (bonding + lone). (3) Determine the electron pair geometry. (4) Name the molecular shape based on the positions of atoms only (not lone pairs). The bond angle decreases by ~2.5° for each lone pair replacing a bonding pair (109.5° → 107° → 104.5°).

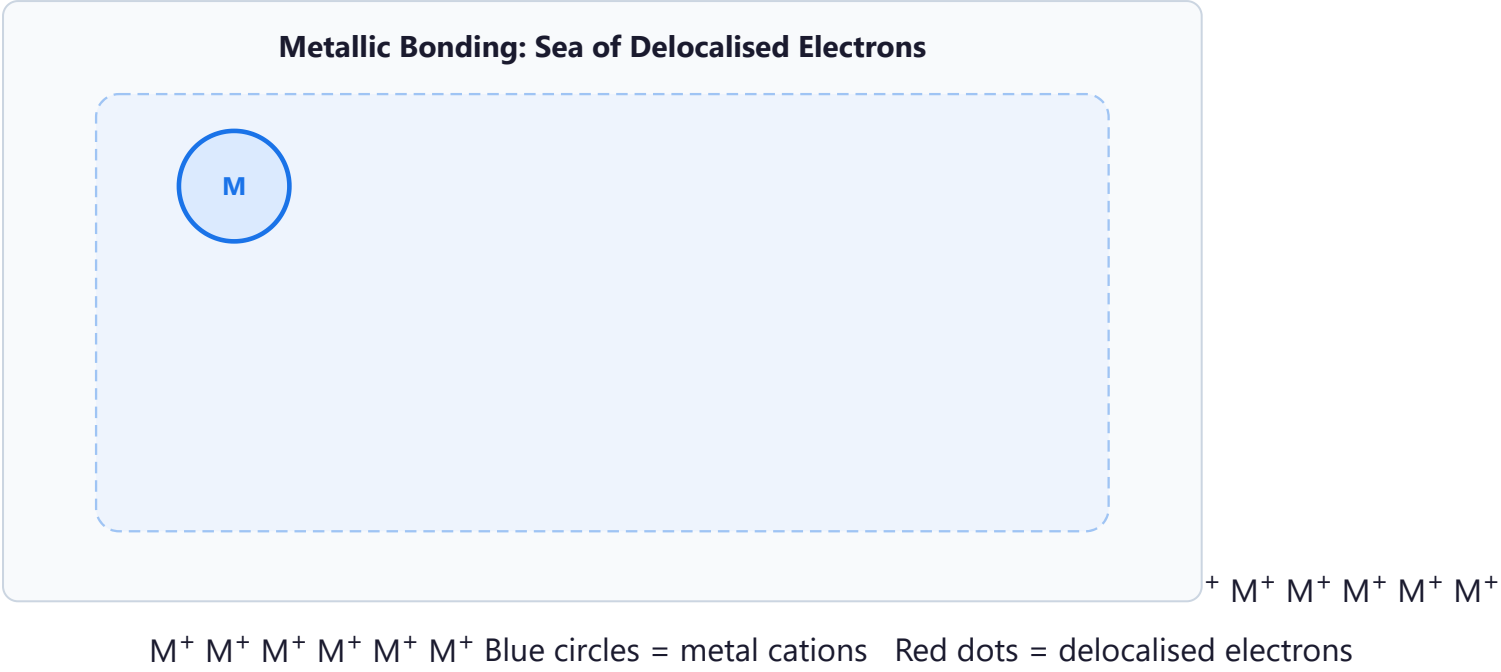
## 1.6 Metallic Bonding

**Metallic bonding:** The strong electrostatic attraction between a lattice of positive metal ions (cations) and a "sea" of delocalised electrons. The outer-shell electrons become detached from individual atoms and are free to move throughout the structure.

**Strength of metallic bonding** depends on:

- **Number of delocalised electrons per atom:** Na (1), Mg (2), Al (3). More delocalised electrons = stronger metallic bond = higher melting point.
- **Ionic radius:** Smaller ions bring the electrons closer to the positive centres, strengthening the attraction.

This explains why melting points increase from Na (98°C) to Mg (649°C) to Al (660°C) across Period 3.




## Properties of Metals

| Property | Explanation |
|---|---|
| Good electrical conductivity | Delocalised electrons flow through the lattice when a potential difference is applied |
| Good thermal conductivity | Delocalised electrons transfer kinetic energy rapidly; lattice vibrations also contribute |
| Malleable and ductile | Layers of ions can slide past each other without breaking bonds — the electron sea adjusts and maintains bonding |
| High melting points (generally) | Strong electrostatic attraction between cations and delocalised electrons throughout the lattice |
| Shiny/lustrous | Delocalised electrons absorb and re-emit light at all visible wavelengths |

# 1.7 Intermolecular Forces

Intermolecular forces are weak attractions **between molecules** (not within molecules). They determine physical properties such as boiling point, solubility, and viscosity. There are three types, in order of

increasing strength:

## 1. London (Dispersion) Forces

Present in ALL molecules and atoms. They arise from temporary, instantaneous dipoles caused by random fluctuations in electron distribution. A temporary dipole in one molecule **induces** a dipole in a neighbouring molecule, creating a weak electrostatic attraction.

**Strength increases with:**

- Number of electrons (larger molecules with greater $M_r$ have more electrons that can be polarised)
- Greater surface area of contact (unbranched chains have greater contact area than branched isomers, giving higher boiling points)

## 2. Permanent Dipole-Dipole Forces

Occur between **polar molecules**. The δ+ end of one molecule is attracted to the δ- end of another molecule. These forces are in addition to London forces and are stronger for molecules of similar size.

## 3. Hydrogen Bonding

> **Hydrogen bond:** A strong type of permanent dipole-dipole interaction that occurs when a hydrogen atom is bonded to a **highly electronegative atom with a lone pair** (N, O, or F). The hydrogen (δ+) is attracted to a lone pair on an N, O, or F atom of a neighbouring molecule. Hydrogen bonds are typically about 1/10th the strength of a covalent bond.

**Requirements for hydrogen bonding:**

1. A hydrogen atom bonded directly to N, O, or F (these atoms are small and very electronegative, making H very δ+)
2. A lone pair on an N, O, or F atom on the neighbouring molecule (to attract the δ+ hydrogen)

**Consequences of hydrogen bonding:**

- Water ($H_2O$), ammonia ($NH_3$), and hydrogen fluoride (HF) have anomalously high boiling points compared to other hydrides in their groups.
- Ice is less dense than liquid water — hydrogen bonds hold molecules in an open, hexagonal lattice with large gaps, so ice floats (crucial for aquatic life).

- Water has a high specific heat capacity (important for climate regulation and biological systems).
- Ethanol is miscible with water (hydrogen bonds between ethanol and water molecules).
- Hydrogen bonding affects protein and DNA structure (the double helix is held together by hydrogen bonds between base pairs).

**Exam Tip:** When explaining boiling points, always: (1) Name the specific intermolecular force. (2) Explain how it arises. (3) Relate it to the energy needed to overcome those forces (not to break covalent bonds — covalent bonds do NOT break during boiling). When comparing boiling points of different molecules, consider the dominant intermolecular force and molecular size/surface area.

## 1.8 Electronegativity and Bond Polarity

**Electronegativity:** The ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself. Measured on the Pauling scale (F = 4.0, most electronegative).

**Trends in electronegativity:**

- **Increases across a period:** Greater nuclear charge with the same shielding pulls bonding electrons more strongly.
- **Decreases down a group:** Greater atomic radius and more shielding reduce the attraction on bonding electrons.
- Fluorine is the most electronegative element (4.0), followed by oxygen (3.5), nitrogen/chlorine (3.0).

**Bond polarity:** When two atoms with different electronegativities form a covalent bond, the bonding electrons are unequally shared, creating a polar bond with partial charges (δ+ on the less electronegative atom, δ- on the more electronegative atom). The greater the electronegativity difference, the more polar the bond.

**Molecular polarity:** A molecule is polar overall if it has polar bonds arranged asymmetrically (so dipoles do not cancel). Examples:

- **Polar molecules:** $H_2O$ (bent, dipoles do not cancel), $NH_3$ (pyramidal), HCl, $CHCl_3$
- **Non-polar molecules with polar bonds:** $CO_2$ (linear, dipoles cancel), $BF_3$ (trigonal planar, dipoles cancel), $CCl_4$ (tetrahedral, dipoles cancel)

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# 1.9 Enthalpy Changes and Calorimetry

**Enthalpy change (ΔH):** The heat energy transferred in a reaction at constant pressure. Measured in kJ $mol^{-1}$.
**Exothermic:** ΔH < 0 (heat released to surroundings, temperature of surroundings rises).
**Endothermic:** ΔH > 0 (heat absorbed from surroundings, temperature of surroundings falls).

## Standard Enthalpy Changes

Standard conditions: 298 K (25°C), 100 kPa pressure, all substances in their standard states, solutions at 1.00 mol $dm^{-3}$.

| Symbol | Name | Definition (per mole, standard conditions) |
|---|---|---|
| $\Delta_c H°$ | Combustion | Complete combustion of one mole of a substance in excess oxygen |
| $\Delta_f H°$ | Formation | Formation of one mole of a compound from its elements in their standard states |
| $\Delta_{neut} H°$ | Neutralisation | Reaction of acid and base to form one mole of water |
| $\Delta_{at} H°$ | Atomisation | Formation of one mole of gaseous atoms from the element in its standard state |
| $\Delta_r H°$ | Reaction | Enthalpy change for the molar quantities shown in the equation |

**Note:** $\Delta_f H°$ of any element in its standard state is zero by definition.

## Calorimetry

Heat energy changes are measured experimentally using calorimetry:

$$q = mc\Delta T$$

where q = heat energy transferred (J), m = mass of solution heated (g), c = specific heat capacity of solution (4.18 J $g^{-1}$ $K^{-1}$ for dilute aqueous solutions), ΔT = temperature change (K or °C).

$$\Delta H = -q / n \quad (\text{kJ mol}^{-1})$$

The negative sign accounts for the sign convention: if the solution warms up (q positive), the reaction is exothermic (ΔH negative).

**Sample calculation:** 50.0 $cm^3$ of 1.00 mol $dm^{-3}$ HCl is mixed with 50.0 $cm^3$ of 1.00 mol $dm^{-3}$ NaOH. Temperature rises by 6.8°C.

- q = 100 × 4.18 × 6.8 = 2842 J = 2.842 kJ
- Moles of water formed = 0.0500 mol (limiting reagent)
- $\Delta_{neut}H$ = -2.842 / 0.0500 = -56.8 kJ $mol^{-1}$

**Exam Tip:** Common sources of error in calorimetry: (1) heat loss to surroundings (the biggest source — makes measured ΔH less exothermic than literature value), (2) incomplete combustion, (3) assuming c = 4.18 and density = 1.00 g $cm^{-3}$, (4) heat absorbed by the calorimeter (not the solution). To improve accuracy, use insulated (polystyrene) cups, a lid, and plot a cooling curve to extrapolate the maximum temperature.

## Hess's Law

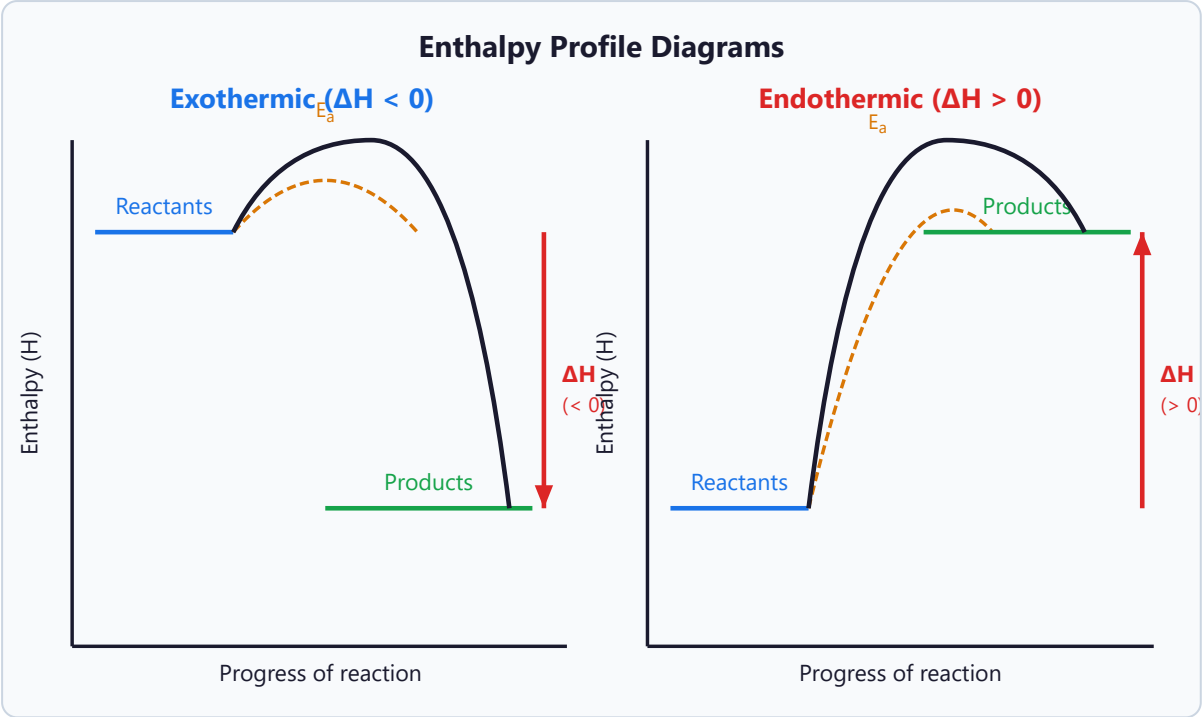
**Hess's Law:** The total enthalpy change for a reaction is independent of the route taken, provided the initial and final conditions are the same. This is a consequence of the law of conservation of energy.

Hess's law allows us to calculate enthalpy changes that cannot be measured directly by constructing enthalpy cycles:

$$\Delta_r H° = \Sigma\Delta_f H°(\text{products}) - \Sigma\Delta_f H°(\text{reactants})$$

$$\Delta_r H° = \Sigma\Delta_c H°(\text{reactants}) - \Sigma\Delta_c H°(\text{products})$$

**Note the reversal of sign:** For formation data, it is products minus reactants. For combustion data, it is reactants minus products. This is because in a Hess cycle, the arrows for combustion go downwards from both reactants and products to the combustion products ($CO_2$ and $H_2O$).




## Bond Enthalpies

**Mean bond enthalpy:** The average energy required to break one mole of a particular type of bond in the gaseous state, averaged across a range of different molecules containing that bond. All species must be in the gaseous state.

Bond enthalpy calculations give approximate values for ΔH because mean bond enthalpies are averages — the actual bond energy depends on the molecular environment. The calculation is:

$$\Delta H = \Sigma(\text{bond enthalpies broken}) - \Sigma(\text{bond enthalpies formed})$$

"Energy in minus energy out" — breaking bonds is endothermic (+), forming bonds is exothermic (-). If more energy is released in forming bonds than is needed to break bonds, the reaction is exothermic overall.

| Bond | Mean Bond Enthalpy / kJ $mol^{-1}$ |
|---|---|
| C-H | 413 |
| C-C | 346 |

| | |
|---|---|
| C=C | 614 |
| C-O | 358 |
| C=O | 805 (in $CO_2$) |
| O-H | 464 |
| O=O | 498 |
| N-H | 391 |
| N≡N | 945 |
| H-H | 436 |
| C-Cl | 338 |
| H-Cl | 432 |

**Exam Tip:** Bond enthalpy calculations only give approximate ΔH values because: (1) mean bond enthalpies are averages over many different compounds, (2) the calculation assumes all species are gaseous (but reactants/products may be liquids or solids). Hess's law calculations using $\Delta_f H°$ or $\Delta_c H°$ data give more accurate results because these values are specific to the compounds involved.